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BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010 **Data Analysis Report**

Project ID: R26-DS-010

Project Title: Multimodal Computational Intelligence for Enhancing Caregiver Performance and Care Delivery – Personalized Sign Language Tutoring System for Caregivers

Introduction

1.1. Background:

Communication barriers between caregivers and deaf or hard-of-hearing patients represent a critical challenge in healthcare settings. Research has documented that 64.4% of deaf patients miss at least half of the important information communicated during medical appointments, while only 32% express satisfaction with healthcare staff communication skills [1]. These barriers directly impact patient safety and healthcare efficiency, with emergency room studies revealing that ASL-using patients experience 9% longer stays than hearing patients an average increase of approximately 30 minutes attributable to communication difficulties [2].

The consequences extend beyond communication inconvenience to compromised care quality and adverse health outcomes. Studies show that deaf patients make less frequent visits to primary care providers while making more trips to emergency rooms, a pattern strongly associated with ineffective communication and limited accessibility of healthcare services [3]. Furthermore, 65% of deaf individuals report difficulty understanding medical instructions, while 66% identify a significant lack of communication tools and support mechanisms within healthcare environments [4].

Despite approximately 70 million people worldwide using sign language as their primary communication method [5], healthcare worker training in sign language remains severely inadequate. A study involving 238 healthcare providers reported that only 6.3% had received formal sign language training, even though 69.3% acknowledged sign language knowledge as “very important” for effective patient care [6]. This training gap persists despite evidence that direct sign language communication improves patient satisfaction, accessibility to preventive care, and overall healthcare outcomes [3].

Existing sign language learning systems primarily emphasize general vocabulary recognition and classification accuracy, with limited focus on caregiver-specific communication contexts, adaptive tutoring, or interpretable corrective feedback. Traditional tutoring approaches lack real-time personalized guidance, making it difficult for learners to identify and correct gestural inaccuracies effectively. Furthermore, many AI-powered sign recognition systems

conflate execution speed with gesture accuracy, penalizing learners who perform gestures correctly but at slower signing speeds.

Research on Dynamic Time Warping (DTW) has demonstrated its effectiveness in handling temporal variability and achieving consistent recognition performance across different gesture execution speeds [7]. However, its integration within intelligent caregiver-focused tutoring systems remains largely unexplored, particularly in combination with explainable feedback mechanisms and adaptive learning strategies.

Recent advances in deep learning and computer vision have significantly improved isolated sign language recognition through temporal-spatial modeling techniques. Architectures such as Bidirectional Long Short-Term Memory (BiLSTM), Gated Recurrent Units (GRU), and hybrid convolutional-recurrent neural networks have demonstrated promising capabilities in capturing dynamic gesture patterns from skeletal landmark sequences extracted from video data. Similarly, landmark-based pose estimation frameworks such as MediaPipe enable efficient extraction of spatial hand and body representations suitable for real-time gesture analysis.

Nevertheless, current research predominantly focuses on benchmark-based sign classification tasks rather than practical caregiver-oriented learning environments. Therefore, there remains a significant research gap in intelligent ASL tutoring systems that simultaneously support caregiver-specific vocabulary learning, speed-tolerant gesture evaluation, adaptive learning pathways, and interpretable corrective feedback within a unified framework.

This research addresses these limitations by developing an AI-assisted personalized tutoring system specifically designed for caregiver-relevant American Sign Language (ASL) vocabulary. The proposed system integrates computer vision-based gesture recognition, temporal sequence modeling, speed-tolerant evaluation using Dynamic Time Warping (DTW) to distinguish execution speed from gestural accuracy, and explainable AI-driven feedback mechanisms to support personalized sign language learning for caregivers. In addition to gesture recognition and corrective feedback, the system incorporates adaptive and gamified lesson progression strategies designed to improve learner engagement, motivation, and long-term retention of caregiver-relevant ASL vocabulary.

1.2. Research Problem:

The primary research problem addressed in this study is the lack of accessible, personalized, and speed-tolerant sign language tutoring systems specifically designed for caregiver communication with deaf or hard-of-hearing individuals. Although recent advances in artificial intelligence and computer vision have improved isolated sign language recognition [8], existing systems remain limited in supporting practical healthcare-oriented learning environments.

Most current sign language learning platforms primarily focus on general vocabulary recognition and classification accuracy rather than adaptive educational support. Existing systems provide limited interpretable feedback, making it difficult for learners to understand how gestures should be corrected in terms of hand positioning, motion trajectories, and gesture timing [9], [10]. Furthermore, many recognition systems fail to accommodate natural variations in signing speed, often penalizing learners who perform gestures correctly but more slowly than expected [11], [12].

In addition, current solutions rarely address caregiver-specific communication scenarios involving emergency assistance, medical symptoms, emotional responses, and daily care interactions. Existing platforms also provide limited personalization capabilities, restricting their ability to adapt learning pathways according to individual learner progress [8].

Specifically, existing systems fail to:

- Distinguish between execution speed and gesture accuracy, leading to incorrect feedback for slow but correctly performed signs
- Provide interpretable corrective feedback explaining which hand movements or gesture trajectories require improvement [9], [10]
- Focus on caregiver-relevant ASL vocabulary across healthcare and daily communication contexts
- Support adaptive learning pathways based on individual learner performance

Therefore, there remains a significant need for an intelligent ASL tutoring framework capable of combining temporal-spatial gesture recognition, speed-tolerant evaluation using Dynamic Time Warping (DTW) [7], explainable feedback generation, and adaptive learning support within a caregiver-focused communication context.

1.3. Objectives:

It Number	Objective	Objective number
IT22289384	Develop MediaPipe-based landmark extraction pipeline for caregiver-specific ASL gesture recognition	1
IT22289384	Implement speed-tolerant gesture evaluation using Dynamic Time Warping (DTW) to distinguish execution speed from accuracy	2
IT22289384	Train and compare deep learning models (BiLSTM, GRU, CNN+BiLSTM) for temporal gesture pattern recognition	3
IT22289384	Design explainable AI feedback system that generates interpretable corrective guidance based on hand position, trajectory, and motion analysis	4

2. Data Exploration

2.1. Data Collection

Data collection followed a multi-source validation strategy to ensure vocabulary relevance, signer diversity, and dataset quality. The vocabulary selection process was conducted in consultation with the Colombo Caregiver and Health Science School (CCHS) to validate the inclusion of caregiver-relevant American Sign Language (ASL) vocabulary.

A total of 66 signs were selected across seven caregiving-oriented categories: Emergency, Health/Medical, Body Parts, Greetings, Emotions/Responses, Food/Drink, and Daily Actions. The selected vocabulary was designed to support common healthcare and caregiving communication scenarios, including emergency assistance, symptom reporting, emotional expression, and daily care interactions.

Video data was collected from two complementary public ASL datasets: the Word-Level American Sign Language (WLASL) dataset and ASL-Citizen. WLASL provided broad vocabulary coverage from 119 signers, while ASL-Citizen contributed additional signer diversity across varied demographic backgrounds and recording environments. Combining both datasets helped reduce the limitations associated with single-source datasets and improved representation of real-world signing variability.

2.2. Dataset Description:

The dataset used in this research was compiled from two publicly available American Sign Language (ASL) datasets: WLASL and ASL-Citizen. The combined dataset was designed to support caregiver-oriented isolated sign recognition across 66 selected ASL classes. The dataset includes signer diversity, varying signing speeds, and multiple recording environments to improve model generalization under real-world conditions.

Data Source	Description	Resource	Size	Key Attributes
WLASL (Li et al., 2019)	Word-level ASL dataset with foundational vocabulary coverage from diverse online sources	GitHub repository	1,350 videos	66 caregiver-relevant signs, 119 signers, RGB videos, variable duration
ASL-Citizen (Desai et al., 2023)	Community-sourced ASL dataset with documented signer diversity and controlled recording conditions	Microsoft Research	1,346 videos	66 signs, 52 documented signers, standardized recording protocol, demographic diversity
Combined Dataset	Merged multi-source dataset following research precedent (WLASL-LEX, 2022)	Custom compilation	2,696 videos	66 caregiver-specific signs, 171+ signers, 7 semantic categories

Each video sample was processed using MediaPipe Holistic landmark extraction to obtain spatial-temporal skeletal representations consisting of hand, pose, and upper-body landmarks. After preprocessing, each gesture sequence was standardized into a fixed-length temporal representation of 40 frames with 774 extracted features per frame. The final model-ready dataset therefore consisted of input tensors with shape: (2684, 40, 774)
where:

- 2684 represents the number of processed gesture samples,
- 40 represents the normalized temporal sequence length,
- 774 represents the extracted landmark and engineered motion features.

The dataset demonstrated moderate class imbalance, with class frequencies ranging from 30 to 93 samples per sign category. To ensure robust model evaluation, stratified train-validation-test splitting and macro-averaged evaluation metrics were used during experimentation.

2.3. Suitability Analysis

2.3.1. Relevance to Individual Research Objectives:

The selected datasets demonstrate strong alignment with the research objectives by supporting gesture recognition, speed-tolerant evaluation, temporal sequence modeling, and explainable feedback generation.

Data Source	Objective 1	Objective 2	Objective 3	Objective 4
WLASL	RGB videos were successfully processed using MediaPipe landmark extraction, achieving 95.8% extraction success and generating standardized landmark sequences suitable for gesture analysis.	Multi-source signer recordings introduced natural temporal variability suitable for DTW-based speed-tolerant evaluation, although signing speed distributions were not explicitly measured.	Temporal sequences standardized to 40 frames enabled successful training of BiLSTM, GRU, and CNN+BiLSTM architectures, achieving baseline temporal recognition performance.	Landmark trajectories enabled spatial motion analysis and DTW-based gesture comparison, though inconsistent hand visibility limited feedback reliability in some samples.
ASL-Citizen	Standardized recording conditions improved landmark extraction consistency, achieving lower preprocessing error rates and reliable fixed-length landmark representations.	Multi-signer diversity and repeated participant recordings introduced realistic signing variability suitable for evaluating speed-tolerant gesture assessment using DTW.	High-quality temporal sequences and diverse signer samples improved deep learning model robustness, contributing to classification performance exceeding 67% test accuracy.	Consistent hand visibility and stabilized landmark trajectories strongly supported explainable feedback generation, gesture decomposition, and adaptive tutoring evaluation.

3. Methodology

3.1. Data Preprocessing:

The preprocessing methodology transforms raw ASL videos into standardized landmark-based feature representations suitable for deep learning model training and DTW-based evaluation. The pipeline consists of six systematic stages: dataset organization, landmark extraction, stabilization, normalization, feature engineering, and train-validation-test splitting.

	Transformation technique			
Data Source	**Objective 1**	**Objective 2**	**Objective 3**	**Objective 4**
WLASL	Support MediaPipe-based landmark extraction from RGB sign language videos	Support DTW-based speed-tolerant gesture evaluation through natural temporal variability	Support temporal deep learning model training using standardized gesture sequences	Support trajectory-based motion analysis for explainable feedback generation
ASL-Citizen	Improve landmark extraction consistency through standardized recording conditions	Support signer-diverse DTW evaluation across varied signing styles	Improve temporal model robustness and generalization through high-quality multi-signer samples	Support interpretable corrective feedback using stable landmark trajectories and consistent hand visibility

3.2. Scalability

The dataset size of 2,684 videos with 40.7 average samples per class is comparable to research-standard isolated sign language recognition datasets. For reference, WLASL-100 contains approximately 20–50 examples per sign, while ASL-Citizen provides similar sample densities. The current dataset scale is sufficient for baseline model development and preliminary DTW evaluation testing.

However, scalability challenges exist for several low-sample classes (excuse: 30, fire: 32, hurt: 32) where model generalization may be limited. The preprocessing pipeline was designed with scalability in mind: batch processing architecture allows incremental dataset expansion, MediaPipe extraction processes videos independently enabling parallel processing, landmark files are stored as compressed. npy arrays minimizing storage overhead (2,684 videos × 40 frames × 774 features × 4 bytes ≈ 332 MB total).

Future scalability improvements could include augmenting low-sample classes through temporal jittering, horizontal flipping, and signing speed variations, expanding to multi-word phrase recognition by concatenating single-sign representations, and integrating continuous signing video segmentation for real-world caregiver interaction scenarios.

3.3. Feature extraction

Feature extraction followed a hierarchical approach prioritizing features critical for caregiver-specific sign recognition:

1. Primary Spatial Features (258 dimensions): MediaPipe pose landmarks (33 × 4 = 132 features) capturing body posture, torso orientation, and shoulder-arm positioning; MediaPipe hand landmarks (21 × 3 per hand = 126 features) encoding finger configurations, hand shapes, and palm orientations.

2. Temporal Motion Features (516 dimensions): First-order velocity (258 features) representing movement direction and speed; Second-order acceleration (258 features) capturing gesture dynamics, rhythm, and smoothness.

3. Derived Features for DTW Evaluation: Hand-specific motion features (126 velocity features) isolating hand movement patterns for speed-tolerant comparison; Trajectory smoothness metrics computed from acceleration magnitude for gesture quality assessment.

Feature validation was performed through statistical analysis of feature distributions across classes, correlation analysis to identify redundant features, and ablation studies comparing position-only versus position + motion feature sets. The motion-enriched 774-feature representation demonstrated superior classification performance over position-only features, justifying the additional computational cost during feature engineering.

4. Modelling and Results

Three deep learning architectures were implemented and compared: BiLSTM (Bidirectional Long Short-Term Memory), GRU (Gated Recurrent Unit), and CNN+BiLSTM hybrid model. All models were trained using the enhanced 774-feature representation with class-balanced loss weighting to address sample imbalance.

BiLSTM Baseline Model: The baseline architecture consisted of two bidirectional LSTM layers (128 units each) with 0.3 dropout, followed by dense layers for 66-class classification. The model learned temporal patterns bidirectionally, capturing both forward and backward gesture progression. Training employed Adam optimizer (learning rate 0.001), categorical cross-entropy loss with class weights, and early stopping with 10-epoch patience. The model converged after 35 epochs with stable validation performance.

GRU Alternative Model: The GRU architecture replaced LSTM cells with simpler gated units, reducing parameter count while maintaining temporal modeling capability. Two bidirectional GRU layers (128 units each) were configured with identical dropout and training parameters as the BiLSTM baseline. GRU training converged faster (28 epochs) with slightly lower computational cost per epoch.

CNN+BiLSTM Hybrid Model: This architecture combined spatial feature extraction via 1D convolutions with temporal sequence modeling. The hybrid design processed the 774-feature input through convolutional layers to extract local motion patterns before feeding the resulting features to bidirectional LSTM layers for temporal integration. This approach aimed to capture both fine-grained spatial relationships and long-range temporal dependencies.

4.1. Key Insights:

The CNN+BiLSTM hybrid model demonstrated superior performance (67.18% accuracy, 63.72% macro F1) compared to BiLSTM baseline (61.58%, 60.49% F1) and GRU (60.27%,59.13% F1). The 5.6% accuracy improvement demonstrates the value of spatial feature extraction via 1D convolutions before temporal modeling with BiLSTM layers.

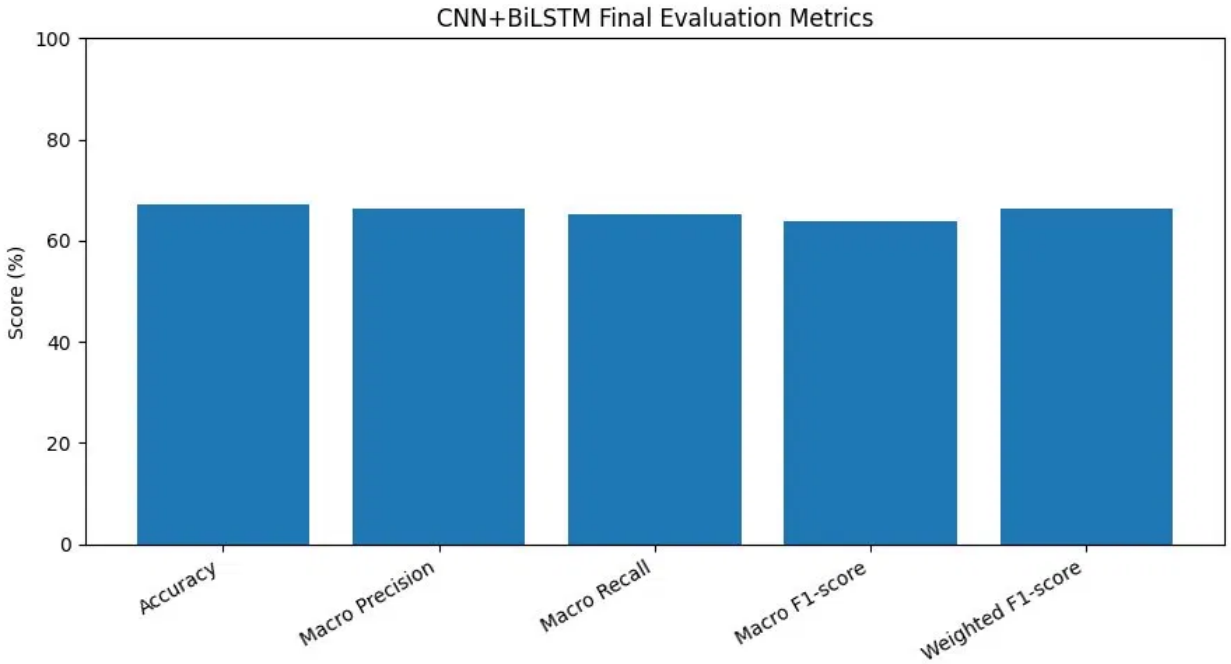


Figure 1. CNN+BiLSTM Final Evaluation Metrics showing 67.18% accuracy and 63.72% macro F1-score on 66-class caregiver ASL recognition task.

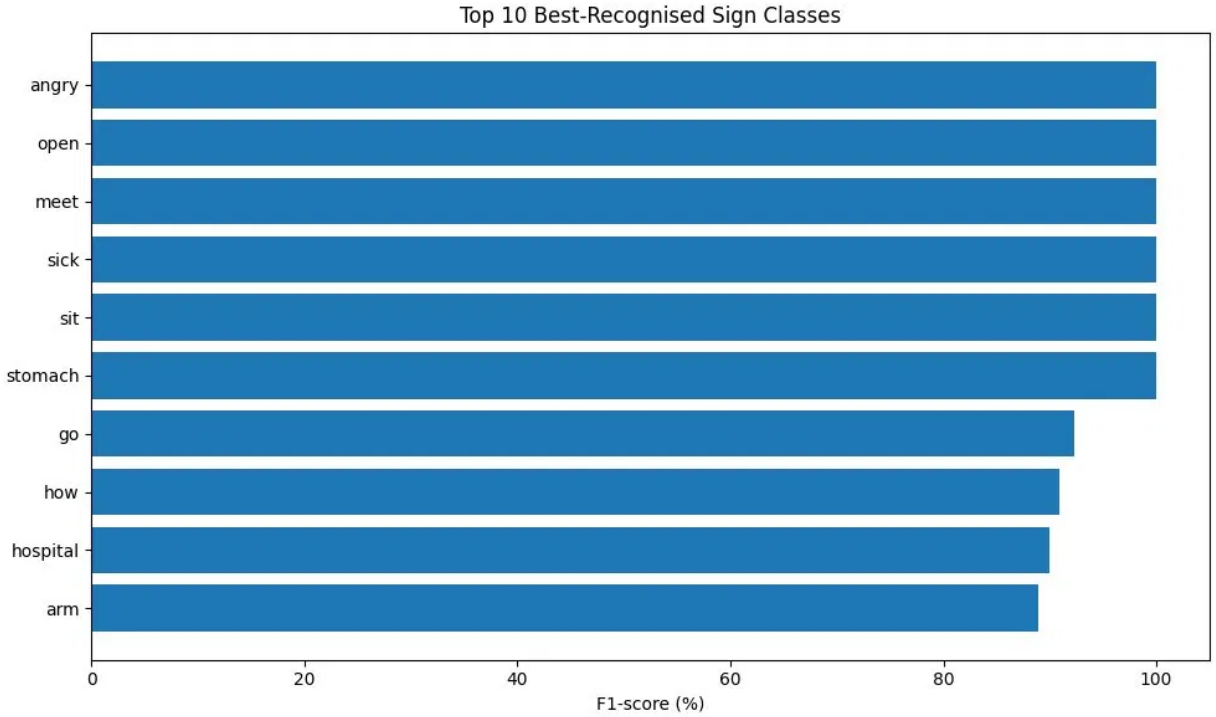


Figure 2. Top 10 best-recognized sign classes by F1-score. Perfect recognition (100%) achieved for angry, open, meet, sick, sit, stomach due to distinctive hand shapes and motion patterns.

Signs achieving perfect or near-perfect recognition (>88% F1) shared common characteristics: distinctive hand configurations (e.g., angry - clenched fist motion), large spatial movement distinguishable from other signs, and minimal visual similarity to other vocabulary items. The model successfully discriminated against signs with clear geometric or motion differences.

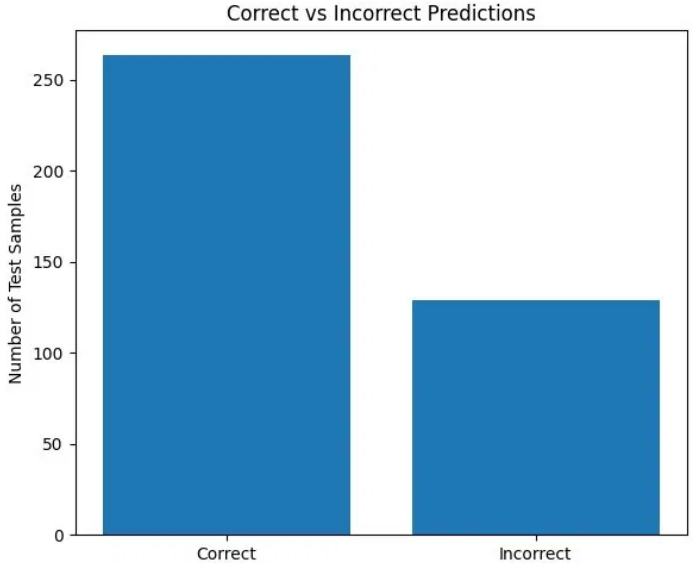


Figure 3. Overall breakdown of correct (264) versus incorrect (129) predictions on 393-sample test set, corresponding to 67.18% accuracy.

The corrective feedback system successfully decomposed gesture errors into interpretable components: Hand Position Error quantifying spatial deviation from reference hand positions, Trajectory Smoothness measuring acceleration variability to detect jerky movements, and Timing Mismatch identifying synchronization issues between learner and reference gesture. Example feedback for a 'help' sign attempt: 'Your right hand started too low. Try raising your starting position by 8cm. Your hand movement was slightly rushed in the middle section. Focus on smooth, controlled motion throughout the sign. Overall timing was good, matching the reference pace.'

4.2. Challenges Faced During Data Analysis:

1. Missing Hand Detections: MediaPipe failed to detect hands in 23.7% of frames due to occlusion, motion blur, or hands positioned outside the camera frame. Forward-fill stabilization reduced this to 8.2% but required careful tuning to avoid over-smoothing genuine hand disappearance events.

2. Class Imbalance: Sample counts varied from 30 (excuse) to 93 (arm) per class, requiring class-weighted loss functions during training to prevent model bias toward high-sample classes.

3. Computational Constraints: Processing 2,684 videos through MediaPipe landmark extraction required approximately 8 hours on Google Colab GPU instances. Parallel processing was not feasible due to Colab's single-instance limitations.

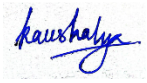
4. Temporal Sequence Variability: Sign duration ranged from 26 to 214 frames, requiring adaptive resampling strategies that balanced temporal resolution preservation with computational efficiency.

5. AI Disclosure

I hereby declare that:

1. I have read and understood the AI Usage Policy for the Research Project (IT4010).
2. I acknowledge that AI tools are to be used only as support tools, not as substitutes for my own understanding and work.
3. I agree to use AI tools responsibly, ethically, and transparently.
4. I will clearly disclose any use of AI tools in my submissions.
5. I understand that I bear full responsibility and accountability for any content, code, model, or output generated using AI tools.
6. I confirm that I will be able to explain, justify, and defend all aspects of my work during evaluations.
7. I understand that failure to comply with this policy may result in academic penalties.

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